

Quotients with positive limiting denominator converge in distribution

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2026-09-07

Abstract

Astra #13 unit u157. If $(U_n, V_n) \Rightarrow (U_0, V_0)$ in $\mathbb{R}^2$ and $V_0 > 0$ almost surely, then $U_n/V_n \Rightarrow U_0/V_0$ (`tendstoInDistribution_div_of_pos`), with Lean's totalised division and no finite- n nonvanishing hypothesis. Division is not continuous at $v = 0$; we divide instead by $\max(v, c)$, which is continuous and agrees with division on $\{v \geq c\}$. By portmanteau on the closed half-line $\{v \leq c\}$ the disagreement probability is eventually at most $\mu'\{V_0 \leq c\}$, which is small for small c ; the approximation lemma of unit 156 finishes. Zero `sorry/axiom`.

1 The things formalised

```
def clipDiv (c : ℝ) (p : ℝ × ℝ) : ℝ := p.1 / max p.2 c
theorem continuous_clipDiv (c) (hc : 0 < c) : Continuous (clipDiv c)
theorem clipDiv_eq (c) (p) (h : c ≤ p.2) : clipDiv c p = p.1 / p.2
theorem tendsto_measure_le_inv_atTop (V : Ω' → ℝ) (hV : Measurable V)
  (hpos : 0 < V) :
  Tendsto (fun m : ℕ => ' { | V ≤ 1 / (m + 1) }) atTop ( 0)
theorem tendstoInDistribution_div_of_pos [l.IsCountablyGenerated] (U V : Ω → ℝ)
  (U V : Ω' → ℝ)
  (hU hV hU hV : measurability)
  (hUV : TendstoInDistribution (fun n => (U n , V n )) l (fun _ => (U , V )))
  (fun _ => ) '
  (hpos : 0 < V) :
  TendstoInDistribution (fun n => U n / V n ) l (fun _ => U / V ) (fun _ => ) '

```

2 Mathematics

Given $\eta > 0$ choose $c = 1/(m + 1)$ with $\mu'\{V_0 \leq c\} < \eta/2$ (the sets $\{V_0 \leq 1/(m + 1)\}$ decrease to $\{V_0 \leq 0\}$, a null set). The marginal $V_n \Rightarrow V_0$ follows from the joint convergence by projection, and portmanteau on the closed set $(-\infty, c]$ gives $\limsup_n \mu\{V_n \leq c\} \leq \mu'\{V_0 \leq c\} < \eta$, hence eventually $\mu\{V_n \leq c\} \leq \eta$. On $\{V_n > c\}$ the clipped quotient equals the true quotient, so the two disagree with probability at most η ; the same holds for the limits. Since the clipped quotients converge by the continuous mapping theorem, unit 156's approximation lemma yields convergence of the quotients.

3 Paper-facing meaning

This is the division step of `cor:empirical_expectation` at chart level: the posterior expectation $Z_n[\varphi]/Z_n$ is the quotient of two jointly convergent normalised chart integrals whose limiting denominator is a.s. positive (unit 154). Astra #13 recommended exactly this route (clipped division plus portmanteau on the denominator marginal) over a general a.e.-continuous mapping theorem or a ball-localisation of the input, and noted that Lean's totalised division removes any need for finite-sample nonvanishing of the chart normaliser.

4 Lean notes

- `continuous_fst.div (continuous_snd.max continuous_const)` with the nonvanishing witness `(lt_max_of_lt_right hc).ne'`.
- `exists_nat_one_div_lt` produces m with $1/(m+1) < x$ for $x > 0$; the deprecated `push_neg` is now `push Not`; for `ENNReal`, `zero_le` takes its argument implicitly (`zero_le _` fails).
- The portmanteau bookkeeping is the same as in unit 153: `ProbabilityMeasure.coe_mk`, `Measure.map_apply_of_aemeasurable` through a universally quantified `have`, and `eventually_lt_of_limsup_lt`.